

REPORT

1. Problem Statement

Industrial sectors are major contributors to global carbon emissions. Small and medium industries lack real-time CO2 tracking due to high costs and complexity. Most factories only track electricity, not direct carbon emissions, leading to undetected inefficiencies and missed optimization opportunities.

2. Proposed Solution

Use AI/ML on existing three-phase energy meter data (voltage, current, power factor) to estimate CO2 emissions without extra sensors. Enables real-time emission tracking, anomaly detection, and actionable optimization suggestions via dashboard visualization.

Data Sources

Data Source	Voltage	Description
(V) Current		Load across A, B, C phases Energy
(A) Power Factor		draw and machine load Energy
Active Power/Energy		efficiency indicator Core input for
Timestamp		CO2 estimation Time-based

4. AI/ML Techniques

Approach	Linear	Use Case	Basic	Benefit	Simple & fast	Handles
Regression		prediction	Non-linear	multiple variables	Predicts	
Random Forest		prediction		future CO2 patterns	Flags	
LSTM / ARIMA		Forecasting & trends		unusual spikes		
Anomaly Detection		Detects inefficiency				

5 Optimization & Insights

Detect energy inefficiencies and provide actionable steps to reduce CO2 emissions.

Issue	Suggested Action	Expected CO2 Reduction
High CO2 in one phase	Redistribute load	5–10%
Idle machines	Auto shutdown	2–5%
Peak hour usage	Shift to off-peak hours	3–8%

6 CO2 Capture and Reuse (Dry Ice Production)

Captured CO₂ can be purified, compressed, and converted into dry ice for industrial use, supporting a circular carbon economy.

Benefits: Reduces greenhouse gas emissions, supports carbon recycling, creates useful products for logistics and pharma sectors.

7. Workflow Diagram (Conceptual)

Three-Phase Energy Meter → Data Preprocessing → AI/ML Model → Alerts & Optimization → Dashboard Visualization

DRAWBACKS:

- 1.High setup cost--> Expensive equipment and technology
- 2.High energy use--> Reduces net climate benefit
- 3.Purity issues--> Requires complex purification
- 4.Market limits--> Dry ice demand not large enough
- 5.Not permanent-->Dry ice re-releases CO₂
- 6.Safety risks--> CO₂ handling requires training
- 7.Economic risk--> Prices & demand unstable
- 8.Policy gaps

OVERCOME IDEAS:

- Capture CO₂ from cleaner sources like fermentation, instead of smokestacks.
Use captured CO₂ to produce multiple products — dry ice, carbonated
 - drinks, fertilizers, fire extinguishers, or synthetic fuels.
 - ◊ Sell CO₂ for greenhouse enrichment in agriculture. ◊
- Partner with logistics, pharma, and food sectors to guarantee demand.

CONCLUSION:

This project transforms standard factory meters into smart carbon monitors using AI/ML. It offers real-time CO₂ estimation, inefficiency detection, and optimization — scalable, cost-effective, and ideal for hackathon demonstrations.